

Where, What, Why, and Importance: Structured Defect Grounding for Text-to-Image Feedback

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Abstract

Despite generating increasingly photorealistic images, text-to-image (T2I) models still exhibit localized, subtle, and structurally complex failures. Diagnosing these failures requires instance-level feedback that answers *where* a defect occurs, *what* type it is, *why* it is defective, and its *importance* to overall image quality. While recent dense-feedback methods move beyond scalar supervision, their heatmap-centric representations still formulate diagnosis as pixel-field regression, making it difficult to localize variable-cardinality defects and bind semantic reasons to individual failures. To address this representation bottleneck, we propose **Structured Defect Grounding (SDG)**, which casts T2I diagnosis as structured set prediction by modeling each defect as a *(location, type, reason, importance)* tuple. To make this formulation trainable and measurable, we introduce **SDG-30K**, a 30K-image dataset with box-grounded annotations across four modern T2I generators, together with a dedicated evaluation protocol, **SDG-Eval**. Building on this structured representation, we further present a diagnosis-to-alignment framework in which a Vision-Language Model (VLM) serves as the SDG detector, and **BoxFlow-GRPO** converts predicted defect sets into box-derived, importance-weighted spatial rewards for diffusion model alignment. Extensive experiments show that our SDG detector outperforms leading proprietary VLMs on structured defect grounding, while SDG-guided rewards consistently improve T2I alignment and support localized image refinement. These results establish SDG as a unified, instance-level interface for diagnosing, evaluating, and enhancing modern generative models.

1 Introduction

Text-to-image (T2I) models [Esser et al., 2024, Betker et al., 2023, Labs, 2024, Cai et al., 2025, Ma et al., 2025a, Team et al., 2025] produce increasingly photorealistic images, yet their failures often remain localized, subtle, and structurally heterogeneous, such as malformed text, implausible geometry, and semantic mismatches. Evaluating such outputs with scalar preference scores [Wu et al., 2023b,a, Xu et al., 2023, Kirstain et al., 2023] collapses these defects into a single global value. Although useful for ranking images, scalar feedback cannot answer the four key questions required for defect diagnosis: *where* the error occurs, *what* type of defect it is, *why* the region is defective, and its *importance* to overall image quality. This limitation reveals a mismatch between global image-level supervision and the localized, instance-level nature of T2I failures.

Recent work attempts to address this need by moving beyond scalar supervision toward dense feedback. RichHF [Liang et al., 2024] introduces artifact and misalignment heatmaps, while ImageDoctor [Guo et al., 2025] augments heatmap-based diagnosis with VLM reasoning. However,

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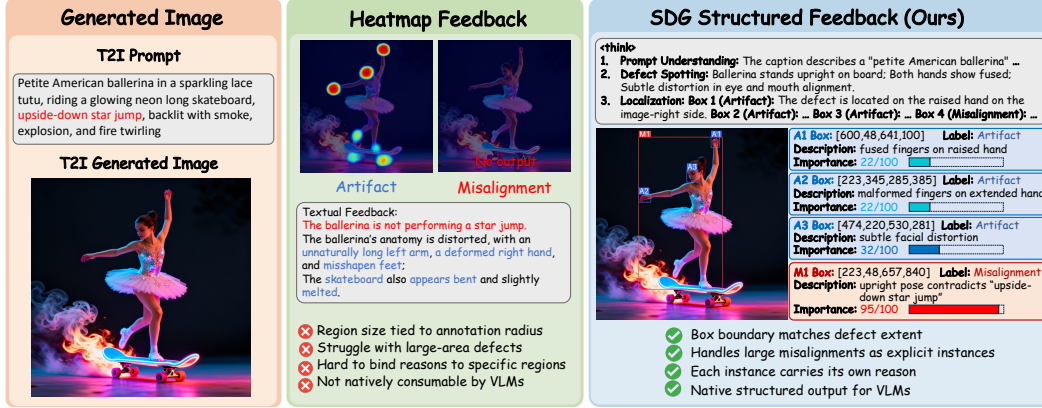


Figure 1: Comparison of dense feedback paradigms for T2I defect diagnosis. **Left:** a T2I generated image with its prompt. **Middle:** heatmap-style feedback produces separate artifact and misalignment maps with textual feedback. **Right:** SDG reasons via a chain-of-thought (CoT) trace, then outputs structured defects with bounding boxes, types, descriptions, and importance scores.

although heatmaps provide dense signals, they still formulate defect diagnosis as pixel-field regression rather than instance-level understanding. As illustrated in Figure 1, this heatmap-centric paradigm exposes representation bottlenecks: spatially, point-derived maps depend on annotator-chosen radii rather than true defect extents [Liang et al., 2024]; semantically, continuous severity fields cannot bind defect types, reasons, or importance scores to individual failures; and architecturally, pixel-level maps are not native outputs of autoregressive VLMs and often require additional decoders or regression heads. These limitations suggest that the key challenge is not merely to make feedback dense, but to represent defects as spatially grounded, semantically explicit, and VLM-compatible instances.

Motivated by this bottleneck, we move from continuous pixel-field regression to structured instance prediction. We introduce **Structured Defect Grounding (SDG)**, which models each defect as a structured (*location, type, reason, importance*) tuple and formulates diagnosis as predicting a variable-length set of such tuples. SDG unifies two defect types, *artifacts* (i.e., image-intrinsic visual flaws) and *misalignments* (i.e., prompt-conditioned semantic errors), within a single instance space, naturally supporting images with multiple heterogeneous defects. To make this formulation trainable and measurable, we construct **SDG-30K**, a 30,096-image dataset with box-grounded artifact and misalignment annotations across four modern T2I generators, and define **SDG-Eval**, a dedicated protocol for structured defect-set evaluation.

Beyond diagnosis, SDG also provides a natural interface for model alignment: boxes define spatial support, defect types and reasons provide semantic diagnoses, and importance scores calibrate reward strength. We convert SDG predictions into box-derived, importance-weighted spatial reward maps and post-train diffusion models with our proposed **BoxFlow-GRPO**, enabling spatially targeted alignment beyond scalar preference optimization. The code, model weights, and dataset are available at <https://github.com/nianbai006/SDG>. Overall, this paper makes following main **contributions**:

- We introduce **Structured Defect Grounding (SDG)**, an instance-level representation that formulates dense T2I diagnosis as variable-cardinality set prediction over structured (*location, type, reason, importance*) tuples.
- We construct **SDG-30K**, a 30,096-image box-grounded defect dataset across four modern T2I generators, and define **SDG-Eval** for image-level and defect-level evaluation.
- We develop a diagnosis-to-alignment framework where a VLM-based **SDG detector** predicts structured defect sets and **BoxFlow-GRPO** converts them into importance-weighted spatial rewards for diffusion model alignment.
- Extensive Experiments demonstrate that our SDG detector outperforms leading proprietary VLMs on structured defect grounding, while SDG-guided rewards improve T2I alignment and support more faithful, actionable image refinement.

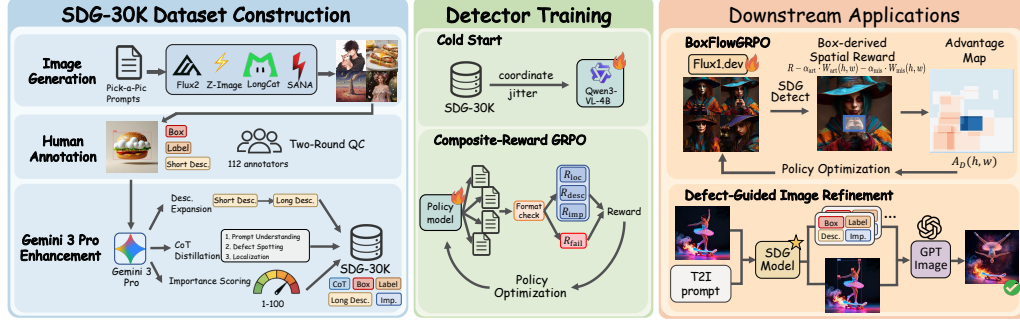


Figure 2: Overview of the SDG framework. **Left:** SDG-30K construction combines human box-level defect annotation across four T2I generators with Gemini 3 Pro enhancement. **Middle:** Two-stage SDG detector training via SFT with coordinate jitter followed by GRPO with a format-gated composite reward. **Right:** Downstream applications. BoxFlow-GRPO converts SDG detections into box-derived spatial rewards for diffusion alignment; defect-guided refinement feeds box overlays and text feedback to GPT-Image-1.5 for image refinement.

2 Related Work

From scalar evaluation to dense T2I feedback. Most T2I evaluators produce scalar scores [Wu et al., 2023b,a, Xu et al., 2023, Kirstain et al., 2023]. RichHF [Liang et al., 2024] introduces heatmap-based dense feedback, ImageDoctor [Guo et al., 2025] predicts heatmaps via a VLM-plus-decoder architecture, and HEIE [Yang et al., 2025] and MagicMirror [Wang et al., 2025a] further enriches feedback with hierarchical explanations and fine-grained artifact taxonomy, respectively. Beyond T2I evaluation, HAD [Wang et al., 2024] and AbHuman [Fang et al., 2024] demonstrate box-level supervision for human-body artifact detection, and LEGION [Kang et al., 2025] combines mask-level localization with natural-language explanations. However, they do not provide a unified instance-level formulation that jointly localizes *artifact* and *misalignment* with localized descriptions.

Structured spatial reasoning in VLMs. Modern VLMs increasingly support explicit spatial structure within an autoregressive generation framework. Qwen2.5-VL [Bai et al., 2025b] supports box- and point-based grounding with structured outputs, while Qwen3-VL [Bai et al., 2025a] further strengthens image-grounded reasoning and spatial understanding. At a finer granularity, SimpleSeg [Song et al., 2026] reformulates segmentation as point-sequence generation entirely in language space. These advances motivate our use of VLMs to generate structured defect instances.

RL for diffusion alignment and image refinement. RL has become a key tool for aligning diffusion models with human preferences. DDPO [Black et al., 2023] frames denoising as a multi-step MDP and applies policy gradients, Diffusion-DPO [Wallace et al., 2024] adapts direct preference optimization to diffusion training, and Flow-GRPO [Liu et al., 2025] applies group relative policy optimization [Shao et al., 2024] to flow-matching models. ImageDoctor [Guo et al., 2025] proposes extending scalar rewards to spatially varying dense reward maps. On the correction side, ReflectionFlow [Zhuo et al., 2025] and HumanRefiner [Fang et al., 2024] demonstrate that defect-guided refinement can improve generation quality. To our knowledge, our work is the first to realize spatially dense advantages in diffusion RL, and further uses structured dense feedback to guide image refinement.

3 SDG-30K: Dataset and Evaluation

3.1 Task Formulation

Given a generated image I and its prompt T , the goal is to predict a variable-cardinality set of structured defect instances $\{(b_i, t_i, r_i, s_i)\}_{i=1}^K$. Here, $K \in \mathbb{N}$ denotes the number of defects in the image. The i -th instance consists of (1) a quantized bounding box $b_i \in \{0, \dots, 1000\}^4$; (2) a defect type $t_i \in \{\text{artifact}, \text{misalignment}\}$; (3) a free-form natural-language description r_i ; and (4) an integer importance score $s_i \in \{1, \dots, 100\}$ reflecting the defect’s perceptual impact on the overall image quality. The detailed definitions and distinctions of these defect types are provided in Appendix A.

Table 1: Comparison with related datasets. SDG-30K is the first to jointly cover *artifact* and *misalignment* within a unified instance-level annotation space with natural-language reasons.

Dataset	Images	Spatial Ann.	Defect Types	NL Reason	Instance-Level	Generators
RichHF-18K [Liang et al., 2024]	18K	Heatmap (point-derived)	Art. + Mis.	×	×	SD-family
HAD [Wang et al., 2024]	37K	Bounding box	Artifact (human body)	×	✓	SDXL, DALL-E, MJ
AbHuman [Fang et al., 2024]	56K	Bounding box	Artifact (human body)	×	✓	SDXL
MagicMirror [Wang et al., 2025a]	343K	Taxonomy label	Artifact	×	×	FLUX, SD3, Kolos, MJ, etc.
LEGION [Kang et al., 2025]	12K	Pixel mask	Artifact	✓	✓	SD, FLUX, GLIDE, etc.
SDG-30K (Ours)	30K	Bounding box	Art. + Mis.	✓	✓	FLUX, Z-Image, LongCat, SANA

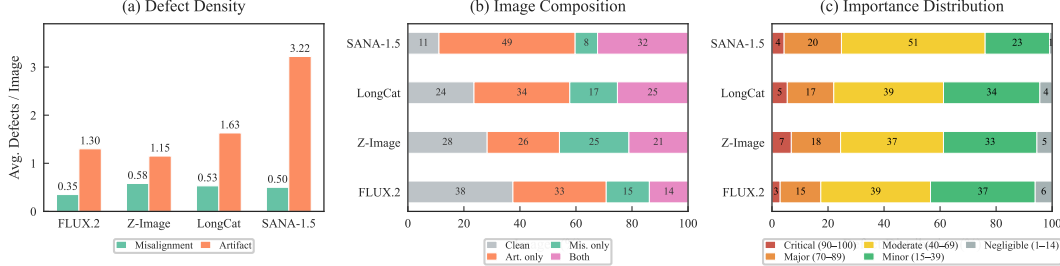


Figure 3: SDG-30K dataset statistics across four generators. (a) Average number of defects per image by type. (b) Image composition showing the proportion of clean, artifact-only, misalignment-only, and both-type images. (c) Distribution of defect importance scores across five severity tiers.

3.2 Evaluation Metrics

We define **SDG-Eval**, an evaluation protocol that scores SDG at both image and defect levels for each defect type $t \in \{\text{artifact}, \text{misalignment}\}$; full definitions are provided in Appendix C. At the image level, **DefTypeF1** measures whether a defect type is present on each test image, while **ClnAcc** reports true-negative accuracy on images with no ground-truth instance of that type ($GT=0$). At the defect level, predictions are matched to ground-truth instances of the same type via class-aware Hungarian matching by IoU. We report **BoxF1@0.1** and **BoxF1@0.5** for localization, **DescCos@0.1** using Qwen3-Embedding-0.6B for description alignment, and **ImpAcc@0.1** as normalized importance accuracy over matched pairs.

3.3 Dataset Construction

Figure 2 (left) illustrates the dataset construction pipeline. SDG-30K contains **30,096** images at 1024×1024 resolution, generated from Pick-a-Pic prompts [Kirstain et al., 2023] using four T2I generators ($\sim 7.8K$ each): FLUX.2-dev [Labs, 2024], Z-Image-Turbo [Cai et al., 2025], LongCat-Image [Ma et al., 2025a], and SANA-1.5-1.6B [Team et al., 2025].

112 annotators ($\sim 1,085$ person-hours) examined each prompt-image pair, drew bounding boxes, assigned top-level labels, and wrote concise Chinese descriptions (≤ 30 characters). Two rounds of review resolved disagreements. To quantify inter-annotator agreement, 16 held-out annotators who did not participate in the original labeling independently re-annotated the test set. The resulting BoxF1@0.5 of 0.278 (*artifact*) and 0.409 (*misalignment*) against the primary annotations serves as a human upper bound for localization (Table 2). Detailed annotation rules are in Appendix A.

Human annotations provide boxes, labels, and short Chinese descriptions but lack reasoning traces and importance scores. We use Gemini 3 Pro [Team et al., 2023] to augment each sample with three components: (1) *description expansion*: from Chinese to detailed English; (2) *reasoning trace distillation*: as a three-step CoT trace (prompt understanding, defect spotting, localization); and (3) *importance scoring*: based on a multi-criteria rubric. Details are in Appendix B.2. Table 1 compares SDG-30K with related datasets.

3.4 Statistics and Splits

Dataset split. The split follows the Pick-a-Pic prompt partition (prompt-disjoint): **28,945** training and **1,151** test images, ensuring that no prompt appears in both splits.

Defect frequency and image composition. Figure 3 summarizes the dataset statistics. *Artifact* instances are more frequent than *misalignment* across all four generators, with SANA-1.5 exhibiting the highest artifact frequency (3.22 per image). Of all images, 25.1% are defect-free, 46.3% artifact-

only, 5.4% misalignment-only, and 23.2% both. This confirms that localized failures remain common in recent T2I generators, motivating dense, instance-level feedback.

Importance distribution. Importance scores estimate each defect’s impact on image quality and prompt faithfulness, enabling failure prioritization and reward weighting. They are centered around the Moderate tier (40–69), mostly between 30 and 80 (Figure 3c), with rare extremes and balanced coverage for training.

4 Structured Defect Grounding Framework

In this section, we instantiate SDG as a diagnosis-to-alignment framework built around the structured-defect representation. We first present the SDG detector, which produces structured defect sets that inherently align with VLM output formats. We then introduce BoxFlow-GRPO to convert SDG’s outputs into box-derived spatial rewards for diffusion alignment.

4.1 SDG Detector

We formulate defect diagnosis as a structured vision-language generation task. Given a generated image and its prompt, SDG first produces a reasoning trace $\mathcal{R}$ and then emits a structured defect set $\mathcal{D}$, where each instance specifies its location, type, reason, and importance.

The SDG detector is trained in two stages (Figure 2, middle). Supervised fine-tuning (SFT) teaches the model to follow the defect-grounding instruction and emit the required structured format, while group relative policy optimization (GRPO) [Shao et al., 2024] further improves localization, description consistency, and importance estimation under a format-validity gate.

Cold Start. We fine-tune Qwen3-VL-4B-Instruct on the SDG-30K training split (Section 3.3). For each image-prompt pair (I, T) , the target sequence concatenates the Gemini-distilled reasoning trace and the structured defect set, denoted as $y = [\mathcal{R}; \mathcal{D}]$. We optimize the model with the standard teacher-forced negative log-likelihood over the target tokens:

$$\mathcal{L}_{\text{SFT}} = - \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid I, T, y_{<t}). \quad (1)$$

To reduce sensitivity to exact coordinate values, we apply *coordinate jitter* to the `<answer>` segment during SFT data loading. For each box $[x_0, y_0, x_1, y_1]$ in normalized $[0, 1000]$ space, each coordinate is independently perturbed by $\delta \sim \mathcal{U}(-10, 10)$, then clamped to $[0, 1000]$ with valid box ordering enforced. The `<think>` segment is kept unchanged because it captures high-level visual reasoning rather than exact numerical coordinates. Since jitter offsets are resampled across epochs, the model observes multiple plausible coordinate variants of the same defect, improving tolerance to minor spatial variation and providing a more robust SFT initialization for subsequent GRPO training.

Composite-Reward GRPO. After SFT, we apply GRPO to directly optimize the structured output using rewards for spatial accuracy, description consistency, and importance estimation. For each prompt, we sample $S=8$ responses, compute the composite reward R_s for each response, and form group-normalized advantages $A_s = (R_s - \bar{R})/\sigma_R$. The policy is then optimized with the clipped GRPO objective:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}[\min(\rho_s A_s, \text{clip}(\rho_s, 1-\epsilon, 1+\epsilon) A_s)] + \beta \text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}}), \quad (2)$$

where $\rho_s = \pi_{\theta}(y_s \mid I, T)/\pi_{\text{old}}(y_s \mid I, T)$ is the importance ratio, π_{old} is the rollout policy, π_{ref} is the fixed reference policy, and $\beta=0.01$.

The composite reward is gated by a format-validity check:

$$R = \begin{cases} \lambda_{\text{loc}} R_{\text{loc}} + \lambda_{\text{desc}} R_{\text{desc}} + \lambda_{\text{imp}} R_{\text{imp}}, & \text{if Format}(y) = \text{true}, \\ R_{\text{fail}}, & \text{otherwise,} \end{cases} \quad (3)$$

where $\lambda_{\text{loc}}, \lambda_{\text{desc}}, \lambda_{\text{imp}} \geq 0$ with $\lambda_{\text{loc}} + \lambda_{\text{desc}} + \lambda_{\text{imp}} = 1$ control the relative importance of each reward component, and $R_{\text{fail}} < 0$ is a fixed penalty for malformed outputs. The format predicate $\text{Format}(y)$ verifies that the response contains well-formed reasoning and answer delimiters, a parseable JSON defect list, and geometrically valid bounding boxes (i.e., $x_0 < x_1$ and $y_0 < y_1$).

The three reward components are defined as follows. R_{loc} measures spatial grounding accuracy: predicted and ground-truth boxes are matched via the Hungarian algorithm with Distance-IoU (DIOU) cost, and the reward reflects match quality with explicit penalties for false negatives and false positives. R_{desc} measures description consistency via embedding cosine similarity (Qwen3-Embedding-0.6B), linearly mapped to $[0, 1]$ and averaged over matched pairs. R_{imp} measures importance estimation accuracy via a clipped absolute-error metric. In practice, we set $\lambda_{\text{loc}}=0.6$, $\lambda_{\text{desc}}=0.25$, $\lambda_{\text{imp}}=0.15$, and $R_{\text{fail}}=-1$. Detailed formulas, boundary-case handling are provided in Appendix D.

4.2 SDG-Guided BoxFlow-GRPO

Once the SDG detector is trained, we translate its dense feedback into spatially varying rewards to guide the reinforcement learning of diffusion models. The closest prior work is ImageDoctor’s DenseFlow-GRPO [Guo et al., 2025], which extends Flow-GRPO [Liu et al., 2025] with a heatmap signal. In its public training code¹, each image is assigned an image-level scalar reward R , normalized across the prompt group into a scalar advantage A ; the predicted heatmap $H \in [0, 1]^{H \times W}$ enters only as a multiplicative mask on the policy-gradient loss, $\mathcal{L} \propto -A \cdot \rho \cdot (1-H)$, where ρ is the importance ratio. The gradient signal is therefore still driven by an image-level scalar, with the heatmap merely down-weighting locations the detector flags as defective rather than contributing a true per-location advantage at the latent grid.

Motivated by this gap, we implement BoxFlow-GRPO around two design choices: box-derived spatial rewards and spatially normalized per-location advantages.

Given a base scalar reward R from any predefined reward model² and the defect bounding boxes detected by our SDG model, we construct a spatially varying reward map in the latent space. For each latent spatial location (h, w) , weighted masks W_{art} , W_{mis} aggregate the predicted importance of all defect boxes covering that location, and the per-location reward subtracts type-specific penalties from the scalar reward:

$$\begin{aligned} W_{\text{type}}(h, w) &= \max_{k \in \mathcal{B}_{\text{type}}(h, w)} \hat{s}_k / 100, \\ \alpha_{\text{art}} &= c_{\text{art}} \cdot \sigma_R^{(\text{group})}, \quad \alpha_{\text{mis}} = c_{\text{mis}} \cdot \sigma_R^{(\text{group})}, \\ R_D(h, w) &= R - \alpha_{\text{art}} W_{\text{art}}(h, w) - \alpha_{\text{mis}} W_{\text{mis}}(h, w). \end{aligned} \quad (4)$$

where $\mathcal{B}_{\text{type}}(h, w)$ denotes the set of boxes of a given type covering location (h, w) and $\hat{s}_k \in \{1, \dots, 100\}$ is the importance predicted by SDG for box k (Section 3.2). We set $W_{\text{type}}(h, w) = 0$ when no box of that type covers (h, w) . We set $c_{\text{art}}=0.5$ and $c_{\text{mis}}=0.05$, making the penalties adaptive to the prompt-group reward standard deviation. This formulation ensures that high-importance defects receive proportionally stronger spatial penalties, while minor defects exert lighter corrections.

After constructing R_D , per-location advantages are computed by normalizing across the K samples within each prompt group at each spatial location:

$$A_D^{(k)}(h, w) = \frac{R_D^{(k)}(h, w) - \mu_D(h, w)}{\sigma_D(h, w) + \epsilon}, \quad (5)$$

where $\mu_D(h, w)$ and $\sigma_D(h, w)$ are the mean and standard deviation of $\{R_D^{(k)}(h, w)\}_{k=1}^K$ computed over the $K=8$ samples in the prompt group, and ϵ is a small constant for numerical stability. Defining the per-location likelihood ratio $\rho_t^{(k)}(h, w) = \pi_\phi(x_{t-1, h, w}^{(k)} | x_t^{(k)}, c) / \pi_{\phi_{\text{old}}}(x_{t-1, h, w}^{(k)} | x_t^{(k)}, c)$, the BoxFlow-GRPO objective is (KL regularization omitted for brevity):

$$\mathcal{J}_{\text{BoxFlow}}(\phi) = \frac{1}{KTHW} \sum_{k, t, h, w} \min\left(\rho_t^{(k)}(h, w) A_D^{(k)}(h, w), \text{clip}\left(\rho_t^{(k)}(h, w), 1-\epsilon, 1+\epsilon\right) A_D^{(k)}(h, w)\right) \quad (6)$$

where T is the number of denoising steps and H, W are the latent spatial dimensions. Compared with a scalar-advantage implementation, this objective preserves spatial variation in both the advantage and the likelihood ratio.

¹<https://github.com/EthanG97/ImageDoctor>

²We use UnifiedReward-2.0 (UR2) [Wang et al., 2025b] in our experiments

5 Experiments

We evaluate SDG from three complementary perspectives: (1) defect grounding quality on SDG-30K, (2) diffusion alignment with structured-feedback rewards via BoxFlow-GRPO, and (3) defect-guided image refinement driven by structured feedback.

5.1 Defect Grounding Results

5.1.1 Setup

Implementation. We fine-tune Qwen3-VL-4B-Instruct on 16 GPUs with DeepSpeed ZeRO-2 for 3 epochs (effective batch size 16, learning rate 3×10^{-5} , cosine schedule). The vision encoder is frozen throughout SFT; Table 4 shows that unfreezing it degrades grounding quality at this dataset scale. For GRPO, we train on 16 GPUs with $S=8$ sampled responses per prompt (temperature 1.0, top- p 0.85) for 2 epochs at learning rate 5×10^{-6} .

Baselines. We compare **SDG** (SFT and GRPO variants) against zero-shot GPT-5.4 and Gemini 3 Pro, both prompted with the same structured output format but without task-specific training. A human reference from 16 independent re-annotators provides a localization upper bound.

5.1.2 Main Results

Table 2: Defect grounding results on SDG-30K test set. Human row: localization upper bound from 16 independent re-annotators. **Bold**: best; underline: second best.

Model	Artifact						Misalignment					
	Image-Level		Defect-Level				Image-Level		Defect-Level			
	All	$GT=0$	$GT>0$				All	$GT=0$	$GT>0$			
	DetTypeF1	ClnAcc	BoxF1@0.1	BoxF1@0.5	DescCos@0.1	ImpAcc@0.1	DetTypeF1	ClnAcc	BoxF1@0.1	BoxF1@0.5	DescCos@0.1	ImpAcc@0.1
GPT-5.4	0.736	0.118	0.237	0.035	0.868	0.812	0.654	0.513	0.457	0.292	0.837	0.820
Gemini 3 Pro	0.733	0.076	0.337	0.200	0.885	0.824	0.696	0.672	0.464	0.307	0.891	0.851
SDG (SFT)	0.776	0.697	0.402	0.255	0.904	0.883	0.636	0.799	0.499	0.376	0.893	0.892
SDG (GRPO)	0.772	0.560	0.404	0.263	0.904	0.887	0.675	0.732	0.511	0.387	0.888	0.893
Human	0.792	0.774	0.462	0.278	–	–	0.712	0.818	0.563	0.409	–	–

Quantitative analysis. Table 2 presents defect grounding results for artifact and misalignment separately. GRPO achieves the strongest BoxF1@0.5 (0.263/0.387) and importance accuracy (0.887/0.893), while SFT remains competitive and attains the highest clean-image accuracy (0.697/0.799). Compared with zero-shot VLM baselines, both SDG variants substantially reduce localization error: GPT-5.4 obtains reasonable image-level artifact detection (F1 0.736) but weak precise artifact localization (BoxF1@0.5 0.035), and Gemini 3 Pro improves localization (0.200/0.307) but remains below SDG. GRPO narrows the gap to the human reference on both artifact (0.263 vs. 0.278) and misalignment (0.387 vs. 0.409), while maintaining high description cosine similarity.

Qualitative comparison. Figure 4 shows qualitative comparisons on SDG-30K across three representative cases: an *artifact-only* image (top), a *misalignment-only* image (middle), and a clean image (bottom). SDG produces instance-level bounding boxes with per-defect descriptions, accurately localizing defects in the first two rows and correctly recognizing the clean case as defect-free. In contrast, ImageDoctor tends to predict heatmap responses on faces and hands even when they are anatomically correct, and struggles to detect prompt-conditioned misalignments.

Cross-dataset generalization. To evaluate generalization, we test SDG on the RichHF-18K test set without any fine-tuning on this dataset, comparing against ImageDoctor [Guo et al., 2025] which is trained on RichHF-18K. Since RichHF-18K provides heatmap annotations, we threshold ImageDoctor heatmaps at two operating points (0.10 and 0.33) for image-level evaluation.

As shown in Table 3, SDG achieves substantially higher misalignment F1 (0.655 vs. 0.250/0.007), demonstrating that structured defect grounding generalizes to unseen data and captures prompt-conditioned misalignments more effectively than heatmap-based methods. ImageDoctor achieves higher artifact F1 at the loose threshold (0.952), as expected from in-domain training, but its misalignment recall is notably poor (0.143/0.004), suggesting that heatmap-based representations struggle with prompt-conditioned misalignment in this setting.



Figure 4: Qualitative comparison on SDG-30K. Rows from top to bottom: artifact-only, misalignment-only, and clean images.

Table 3: Image-level defect detection on RichHF-18K. ImageDoctor is trained in-domain; SDG is zero-shot.

Method	Artifact			Misalignment		
	Prec.	Rec.	F1	Prec.	Rec.	F1
ImageDoctor @0.10	0.965	0.939	0.952	0.983	0.143	0.250
ImageDoctor @0.33	0.977	0.762	0.856	1.000	0.004	0.007
SDG (ours)	0.970	0.698	0.812	0.951	0.500	0.655

Table 4: Key ablation results on SDG-30K. Full metrics are in Table 7.

Variant	Art. BoxF1@0.5	Mis. BoxF1@0.5
SFT	0.255	0.376
GRPO	0.263	0.387
SFT w/o step 1	0.218	0.329
SFT w/o step 3	0.253	0.281
GRPO w/o CoT	0.288	0.352
SFT unfreeze ViT	0.203	0.305
SFT w/o jitter	0.253	0.360

5.1.3 Ablation Study

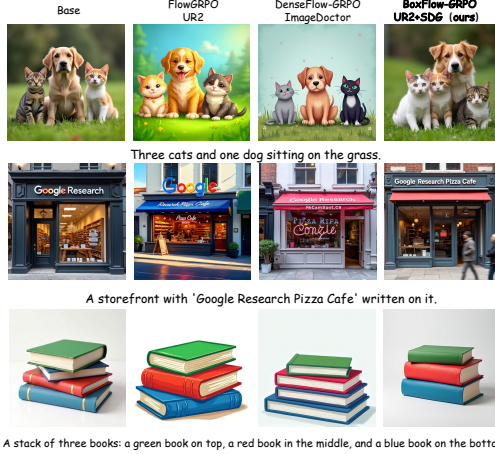
Table 4 summarizes the main ablation trends, with full metrics in Appendix E. GRPO improves localization over SFT (artifact/misalignment BoxF1@0.5: 0.263/0.387 vs. 0.255/0.376), confirming that policy optimization refines spatial precision. Removing CoT steps hurts prompt-conditioned misalignment more than artifact detection, and removing the reasoning trace from GRPO lowers misalignment BoxF1@0.5 from 0.387 to 0.352. Unfreezing the vision encoder substantially degrades localization, while coordinate jitter mainly improves image-level robustness (full results in Table 7).

5.2 Downstream Applications

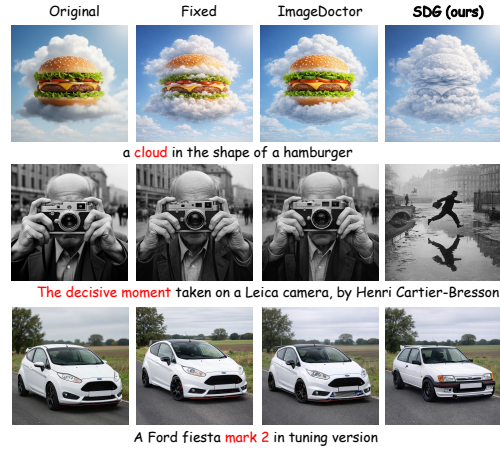
5.2.1 BoxFlow-GRPO

Setup. We apply the dense reward construction of Section 4.2 to FLUX.1-dev, following Flow-GRPO [Liu et al., 2025]. The scalar base reward is UnifiedReward-2.0 (UR2) [Wang et al., 2025b]; SDG detections convert it into a dense reward by subtracting importance-weighted artifact and misalignment penalties at covered latent locations. Training uses Pick-a-Pic [Kirstain et al., 2023] prompts held out from SDG-30K (prompt-disjoint), for 500 optimization steps at 512×512 resolution on 8 GPUs with LoRA (rank 64, $\alpha=128$) and learning rate 3×10^{-4} . We evaluate on DrawBench [Saharia et al., 2022] using PickScore [Kirstain et al., 2023], CLIPScore [Hessel et al., 2021], HPSv3 [Ma et al., 2025b], DeQA [You et al., 2025], and the real-image probability $P(\text{real})$ from Forensic-Chat [Lin et al., 2025], obtained by a 2-class softmax over its “real” vs. “fake” token logits.

Results. As shown in Table 5, baseline RL variants tend to drift toward more illustration- or anime-like outputs after training, a shortcut that can increase reward-model scores without preserving photographic realism. This reward hacking is reflected by the drop in $P(\text{real})$ for all baselines



A stack of three books: a green book on top, a red book in the middle, and a blue book on the bottom.



A Ford fiesta mark 2 in tuning version

Figure 5: Qualitative comparison of BoxFlow-GRPO. Figure 6: Qualitative comparison of defect-guided image refinement via GPT-Image-1.5.

Table 5: Downstream performance of BoxFlow-GRPO and baselines. Baseline RL variants can improve reward-model scores by drifting toward illustration- or anime-like outputs, reflected by lower $P(\text{real})$; **Only** BoxFlow-GRPO improves all five reported dimensions. Parenthesized values are relative change vs. Base (**green**: improvement, **red**: regression); Avg is the mean relative change.

Method	Reward	PickScore	CLIPScore	HPSv3	DeQA	$P(\text{real})$	Avg
Base	–	22.84	0.912	11.75	0.876	0.211	–
Flow-GRPO	UR2	22.78 (-0.3%)	0.908 (-0.4%)	11.27 (-4.1%)	0.882 (+0.7%)	0.149 (-29.4%)	-6.7%
	ImageDoctor	23.17 (+1.4%)	<u>0.937</u> (+2.7%)	12.23 (+4.1%)	0.857 (-2.2%)	<u>0.201</u> (-4.7%)	+0.3%
DenseFlow-GRPO	ImageDoctor	22.88 (+0.2%)	0.945 (+3.6%)	11.57 (-1.5%)	0.864 (-1.4%)	0.199 (-5.7%)	-1.0%
BoxFlow-GRPO (ours)	UR2+SDG	<u>22.91</u> (+0.3%)	0.915 (+0.3%)	<u>12.14</u> (+3.4%)	<u>0.877</u> (+0.1%)	0.228 (+8.1%)	+2.4%

relative to Base. BoxFlow-GRPO instead achieves the best average relative change (+2.4%) and the highest $P(\text{real})$ (0.228, above Base) while maintaining competitive preference and quality metrics. Figure 5 provides a qualitative example where BoxFlow-GRPO reduces both artifacts and prompt misalignment while preserving photographic realism.

5.2.2 Defect-Guided Image Refinement

Setup. SDG first diagnoses defects and provides GPT-Image-1.5 with a box overlay and structured text feedback. We compare it with **Fixed** caption-only editing and **ImageDoctor** heatmap-based feedback, using the same editor for all methods. Two annotators independently and blindly compare paired outputs on 873 valid samples retained after GPT-Image-1.5 filtering, assigning Good/Same/Bad labels following HunyuanImage 3.0 [Cao et al., 2025], where Good means SDG is preferred.

Results. As shown in Table 6, SDG obtains higher Good than Bad rates against both ImageDoctor (11.00% vs. 3.90%) and Fixed (10.31% vs. 2.75%). The high Same rate (85.1–86.9%) reflects the strong GPT-Image-1.5 editor and the already high quality of many inputs. Figure 6 shows that SDG can still enable targeted semantic correction, e.g., identifying that an image depicts a modern Ford Fiesta rather than the prompted Mark 2. Additional qualitative refinement results are provided in Appendix E.5.

Table 6: GSB rates between SDG and baseline methods over 873 valid samples.

Comparison	Good (%)	Same (%)	Bad (%)
SDG vs ImageDoctor	11.00	85.11	3.90
SDG vs Fixed	10.31	86.94	2.75

6 Conclusion

We presented **Structured Defect Grounding (SDG)**, an instance-level formulation for dense text-to-image (T2I) feedback that models each defect as a (*location, type, reason, importance*) tuple and

casts diagnosis as variable-cardinality structured set prediction. This formulation supports **SDG-30K**, a 30K-image box-grounded dataset across four modern T2I generators, and **SDG-Eval** for structured defect-set evaluation. It also enables a diagnosis-to-alignment pipeline in which a VLM-based detector predicts defect sets and **BoxFlow-GRPO** converts them into dense spatial rewards for diffusion alignment. Experiments demonstrate that SDG outperforms leading proprietary VLMs in zero-shot defect grounding, improves T2I alignment, and supports actionable defect-guided image refinement, making it a practical interface for evaluating and improving modern generative models.

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A Annotation Guidelines

This appendix describes the annotation protocol used to construct the SDG-30K dataset. The annotation task was conducted as data curation for generated images rather than as a user study. Annotators were compensated above the local minimum wage in the data-collection location.

A.1 Interface and Workflow

The annotation interface consists of two synchronized panels: a raw image canvas on which annotators draw the final bounding boxes and a metadata panel showing the generation prompt together with annotation controls.

Each image is processed in two passes. In the *initial annotation pass*, annotators examine the prompt-image pair, draw defect boxes from scratch, assign top-level labels, and write concise descriptions for each confirmed defect. In the *global scan pass*, annotators re-examine the full image to identify missed defects, adjust box boundaries, and improve annotation completeness. No machine-generated candidate boxes or defect descriptions are shown during manual annotation.

A.2 Defect-Type Taxonomy

We define two top-level defect categories, each with fine-grained subtypes.

Misalignment captures inconsistencies between the generation prompt and image content. Subtypes include missing objects, extra objects, attribute mismatches (color, count, material), spatial-relation errors, action mismatches, and style mismatches. For example, a prompt requesting “two cats on a red sofa” paired with an image showing three cats constitutes a count-based misalignment.

Artifact captures visual plausibility defects independent of the prompt. Subtypes include anatomical distortions (e.g., malformed hands), geometric deformations, texture abnormalities, edge and contour defects, text garbling, and lighting inconsistencies. For example, fused fingers or impossible joint angles constitute anatomical artifacts regardless of the prompt content.

The taxonomy explicitly separates semantic mismatch from visual corruption: green eyes when blue were requested is misalignment, while geometrically distorted eyes is an artifact. Intentional artistic stylization (e.g., cubist deformation) is not labeled as a defect; only non-intentional structural or perceptual inconsistencies qualify.

A.3 Boxing and Description Principles

Although bounding boxes are coarser than pixel masks, they provide a practical trade-off between spatial specificity, annotation efficiency, and compatibility with VLM outputs. Bounding boxes should tightly cover the defective location while excluding unnecessary background. Each box corresponds to one concrete defect: when a large defect location contains multiple independent issues, annotators split them into separate boxes; when multiple small detections reflect a single coherent failure mode, they may be merged.

For each box, annotators provide a concise reason statement (target length ≤ 30 Chinese characters) that specifically describes the defect. Vague comments such as “looks wrong” are prohibited; descriptions must reference concrete visual evidence (e.g., “left hand has six fingers” or “hat mentioned in prompt is absent”).

A.4 Clean Images

The annotation platform required at least one bounding box per submission. For images without defects, annotators drew a placeholder box and assigned an explicit *no-problem* tag. During post-processing, these samples are converted to empty defect sets, preserving them as valid negative examples for training and evaluation.

A.5 Prompt Interpretation Protocol

To ensure the correctness of prompt interpretation, annotators were instructed to consult search engines whenever a prompt involved unfamiliar named entities, cultural references, artistic styles, products, landmarks, or other domain-specific concepts. External search was used only to clarify the semantic background of the prompt, rather than to provide defect labels directly. This protocol reduces annotation noise caused by incomplete prior knowledge and improves the consistency of prompt-conditioned *misalignment* annotation.

A.6 Placeholder Box Handling

Due to a platform constraint, each sample required at least one submitted box. For truly clean images, annotators drew a placeholder box and assigned an explicit “no-problem” tag. In post-processing, these samples are mapped to empty defect sets so that clean images remain label-consistent during training and evaluation.

A.7 Importance Scoring Rubric

Each defect instance receives an importance score (integer from 1 to 100) reflecting how much it affects overall image quality and caption faithfulness, assessed via a rubric-guided Gemini protocol. The four criteria are considered in order of priority:

1. *Visual prominence* — how easily a typical viewer can spot the defect at normal viewing distance.
2. *Semantic impact* — whether the defect changes the meaning, identity, or key content relative to the prompt.
3. *Area coverage* — larger defects affecting more of the image score higher.
4. *Location* — defects on the main subject or focal area score higher than those in the background.

Scores are grouped into five tiers: Critical (90–100), Major (70–89), Moderate (40–69), Minor (15–39), and Negligible (1–14). Importance annotations are obtained by prompting Gemini with the image, the original prompt, and the human-annotated defect set, and are used both as SFT supervision and as a GRPO reward signal.

B Prompt Templates

This appendix provides the full prompt templates used in SDG. The same user prompt (Appendix B.1) is used for SFT training, GRPO training, and inference. The system prompt uses the default Qwen chat template (“You are a helpful assistant.”).

B.1 SFT / GRPO / Inference User Prompt

SDG User Prompt

```
You are an AI image quality evaluator. You will be given one image to analyze.
### Definitions
Misalignment: Areas where the image content does NOT match the text caption,
including:
- Missing objects: Objects mentioned in caption but not present in image
- Extra objects: Objects present in image but not mentioned in caption
- Wrong attributes: Incorrect color, size, material, count, or other properties
- Wrong spatial relationships: Incorrect positions, orientations, or
arrangements
Artifact: Visual defects in the generated image, including:
- Distorted anatomy: Malformed hands, extra/missing limbs, wrong number of
fingers
- Duplicated/missing parts: Repeated or absent body parts, objects
- Warped geometry: Perspective errors, impossible shapes
- Texture issues: Melted, smeared, or overly smooth textures
- Unnatural edges: Jagged, broken, or blurry boundaries
```


- Garbled text: Unreadable or malformed text/letters
- Lighting inconsistencies: Wrong shadows, reflections, or light sources

Text Caption: {caption}

Goal: Produce a detailed analysis of the image quality and output bounding boxes with severity scores for all detected issues.

Strict Output Rules

Output TWO blocks in this exact order:

- 1) <think> - Your detailed analysis
- 2) <answer> - JSON list of bounding boxes

Think Format (STRICT)

<think>

Step 1: Caption Understanding

- Briefly summarize what the caption requires (subject, key attributes, actions, setting, style/composition if mentioned).

Step 2: Visual Analysis & Defect Spotting (Issue Summary)

- Describe the quality issues you observe in the image.
- You MAY merge similar issues into a single bullet.
- Each bullet MUST include:
 - (a) the issue category (artifact or misalignment, or both if needed)
 - (b) what is affected (object/part)
 - (c) concrete visual evidence (what specifically looks wrong/missing/mismatched)
- Do NOT mention numeric coordinates.

Step 3: Localization (Box-by-Box Grounding)

- Provide a detailed, precise localization statement for EACH defect instance (one line per box).
- Do NOT mention numeric coordinates.
- Each localization line MUST include all of the following information in natural language:
 - 1) Anchor: the exact object/part involved
 - 2) Position: image-based cues (image-left/right, upper/lower, center, near the border)
 - 3) Interaction cue (when applicable): holding/touching/overlapping/merging/etc.
 - 4) Scale description: tiny localized detail, part-sized area, large area, or whole image
 - 5) Shape/orientation: compact, elongated, runs along a boundary, wraps around an object
- Do NOT invent new defects; each line must correspond to exactly one defect instance.

</think>

Answer Format (for <answer>)

```
[{"box_2d": [x0, y0, x1, y1], "label": "misalignment"|"artifact",
"description": "brief description of the issue", "importance": N}]
```

Bounding box coordinates are in normalized 0-1000 space: [x0, y0, x1, y1].
If there are no issues, output an empty list.

Importance Scoring

For EACH box, assign an integer importance score from 1 to 100:

- 90-100: Critical -- immediately obvious, ruins the image.
- 70-89: Major -- clearly visible at normal viewing distance.
- 40-69: Moderate -- noticeable on closer inspection.
- 15-39: Minor -- only visible on careful examination.
- 1-14: Negligible -- barely perceptible.

Description Style Guide

- 18 to 45 words per description.
- Must mention the affected object/part.
- Must include at least one concrete evidence phrase.
- Do NOT use vague words like "weird/strange/bad".
- Do NOT include numeric coordinates in description.

Now analyze the image and produce your output:

B.2 Gemini Distillation Prompt

During data preparation, we use Gemini 3 Pro to translate Chinese defect descriptions into English, expand them with richer detail, and assign importance scores. The model receives the image, caption, and human-annotated defect boxes as hidden context. The full prompt is shown below.

Gemini Distillation Prompt

```
You are an AI image quality evaluator. You will be shown an image and asked to
identify quality issues.
Hidden context (DO NOT reveal):
The human annotations below are Ground Truth. You MUST keep the same number of
boxes, the same coordinates, and the same labels in the final JSON. You must
behave as if you discovered the issues independently from the image and caption.
Ground Truth annotations (DO NOT modify coordinates or labels; you may
rewrite/expand description in English):
- Artifact boxes: {artifact_bbox_text}
- Misalignment boxes: {misalignment_bbox_text}
Text Caption: {caption}
===== GLOBAL CONSTRAINTS (MUST FOLLOW) =====
1) NEVER mention annotations, boxes, ground truth, translation, or any external
hints.
2) Do NOT invent extra defects or remove any defect. The final JSON must contain
exactly the same set of boxes/labels as provided.
3) IMPORTANT: box_2d uses RELATIVE coordinates on a 0-1000 scale.
- Format: [x_min, y_min, x_max, y_max] (xyxy)
- You MUST output these coordinates exactly as given in the final JSON.
4) Do NOT write numeric coordinates in the <think> block.
5) Output MUST contain exactly TWO blocks in this order:
(1) <think> ... </think> (2) <answer> ... </answer>
6) <answer> MUST be a JSON list in xyxy format:
[{"box_2d": [x0,y0,x1,y1], "label": "artifact"|"misalignment",
"description": "...", "importance": N}, ...]
===== IMPORTANCE SCORING (FIELD: "importance") =====
For EACH box, assign an integer importance score from 1 to 100:
- 90-100: Critical -- immediately obvious, ruins the image.
- 70-89: Major -- clearly visible at normal viewing distance.
- 40-69: Moderate -- noticeable on closer inspection.
- 15-39: Minor -- only visible on careful examination.
- 1-14: Negligible -- barely perceptible.
Scoring criteria (in order of priority):
a) Visual prominence -- How easily can a typical viewer spot this defect at
normal viewing distance?
b) Semantic impact -- Does this defect change the meaning, identity, or key
content of the image relative to the caption?
c) Area coverage -- Larger defects affecting more of the image score higher.
d) Location -- Defects on the main subject or focal area score higher than those
in background/periphery.
===== THINKING FORMAT (STRICT) =====
<think>
### Step 1: Caption Understanding
### Step 2: Visual Analysis & Defect Spotting (Issue Summary)
### Step 3: Localization (Box-by-Box Grounding)
</think>
===== DESCRIPTION STYLE GUIDE (FOR <answer>) =====
For EACH box in the final JSON, rewrite description into English with richer
detail:
- 18 to 45 words per description.
- Must mention the affected object/part.
- Must include at least one concrete evidence phrase.
- Do NOT use vague words like "weird/strange/bad".
- Do NOT mention translation or ground truth.
```

```

===== FEW-SHOT EXAMPLES =====
[EXAMPLE A: NON-EMPTY GT]
Caption: a rock musician sticking tongue out holding a sign that says hail Satan
<think>
### Step 1: Caption Understanding
- The caption asks for a rock musician sticking their tongue out while holding a sign reading "hail Satan".
### Step 2: Visual Analysis & Defect Spotting
- Misalignment: The person does not convincingly match the specified celebrity identity.
- Misalignment: The sign text is not rendered as the requested phrase.
- Artifact: Ring edges blend into skin and local finger texture appears unnaturally bumpy.
### Step 3: Localization
- Box 1 (artifact): Ring-skin contact on image-right hand, tiny localized detail.
- Box 2 (artifact): Mid-finger ring edge, compact boundary location.
- Box 3 (artifact): Below rings on image-right fingers, tiny skin patch.
- Box 4 (artifact): Image-left hand ring-to-finger junction, tiny localized.
- Box 5 (misalignment): Upper-central face location, part-sized.
- Box 6 (misalignment): Central sign surface with lettering, part-sized.
</think>
<answer>
[{"box_2d": [777, 604, 796, 627], "label": "artifact",
"description": "Ring appears fused into finger with smeared metal-to-skin boundary...",
"importance": 45}, ...]
</answer>
[EXAMPLE B: EMPTY GT]
Caption: a girl with pink curly hair wearing a witch hat
<think> ... No defects identified. </think>
<answer> [] </answer>
Now analyze the image and respond with exactly TWO blocks (<think> then <answer>).

```

C Evaluation Metric Details

Let $\mathcal{T} = \{\text{artifact}, \text{misalignment}\}$ and let $\mathcal{D}$ be the evaluation set. For image d , denote the ground-truth and predicted defect sets as $G(d) = \{(b_i, t_i, r_i, s_i)\}_{i=1}^{N_d}$ and $P(d) = \{(\hat{b}_j, \hat{t}_j, \hat{r}_j, \hat{s}_j)\}_{j=1}^{M_d}$. For type t , let $G_t(d)$ and $P_t(d)$ be the subsets whose type is t .

Image-level metrics. We define binary indicators $y_{t,d} = \mathbf{1}[|G_t(d)| > 0]$ and $\hat{y}_{t,d} = \mathbf{1}[|P_t(d)| > 0]$. DetTypeF1 is the type-specific image-level F1:

$$Pr_t = \frac{\sum_{d \in \mathcal{D}} y_{t,d} \hat{y}_{t,d}}{\sum_{d \in \mathcal{D}} \hat{y}_{t,d}}, \quad Re_t = \frac{\sum_{d \in \mathcal{D}} y_{t,d} \hat{y}_{t,d}}{\sum_{d \in \mathcal{D}} y_{t,d}}, \quad \text{DetTypeF1}_t = \frac{2Pr_t Re_t}{Pr_t + Re_t}. \quad (7)$$

Clean-image accuracy is the true-negative rate on $\mathcal{D}_t^- = \{d \in \mathcal{D} \mid y_{t,d} = 0\}$:

$$\text{ClnAcc}_t = \frac{1}{|\mathcal{D}_t^-|} \sum_{d \in \mathcal{D}_t^-} \mathbf{1}[\hat{y}_{t,d} = 0]. \quad (8)$$

Defect-level metrics. For each d , we compute a class-aware Hungarian matching [Kuhn, 1955] between $G(d)$ and $P(d)$ under the constraint $t_g = t_p$. Let $\mathcal{M}_{\tau,t}$ be the union of matched pairs of type t whose IoU is at least τ . For $\tau \in \{0.1, 0.5\}$, localization precision, recall, and F1 are

$$\text{BoxPr}_t = \frac{|\mathcal{M}_{\tau,t}|}{\sum_{d \in \mathcal{D}} |P_t(d)|}, \quad \text{BoxRe}_t = \frac{|\mathcal{M}_{\tau,t}|}{\sum_{d \in \mathcal{D}} |G_t(d)|}, \quad \text{BoxF1}_t = \frac{2|\mathcal{M}_{\tau,t}|}{\sum_{d \in \mathcal{D}} (|G_t(d)| + |P_t(d)|)}. \quad (9)$$

For each valid matched pair $(g, p) \in \mathcal{M}_{\tau, t}$, DescCos is the mean cosine similarity between Qwen3-Embedding-0.6B embeddings of r_g and $\hat{r}_p$, and ImpAcc is normalized absolute-error accuracy:

$$\text{DescCos}_t = \frac{1}{|\mathcal{M}_{\tau, t}|} \sum_{(g, p) \in \mathcal{M}_{\tau, t}} \langle r_g, \hat{r}_p \rangle, \quad \text{ImpAcc}_t = \frac{1}{|\mathcal{M}_{\tau, t}|} \sum_{(g, p) \in \mathcal{M}_{\tau, t}} \left(1 - \frac{|s_g - \hat{s}_p|}{100} \right). \quad (10)$$

D Experimental Details

This appendix provides the complete reward computation and training details for the GRPO stage described in Section 4.1.

D.1 Reward Formulation

Composite reward. The composite reward is gated by a format check:

$$R = \begin{cases} 0.6 R_{\text{diou}} + 0.25 R_{\text{desc}} + 0.15 R_{\text{imp}}, & \text{if format valid,} \\ -1, & \text{otherwise.} \end{cases} \quad (11)$$

The format gate verifies that the response contains valid `<think>` tags, a parseable JSON defect list in `<answer>`, and properly ordered xxyy coordinates.

Grounding accuracy (R_{diou}). Predicted and ground-truth boxes are matched using the Hungarian algorithm with DIoU as the cost metric, yielding optimal one-to-one assignments. For each matched pair $(i, j) \in \mathcal{M}$, the spatial reward is the DIoU score of that pair.

Edge cases are handled as follows:

- **Correct rejection** (both sets empty): $R_{\text{diou}} = 0.3$.
- **Miss** (ground-truth defects exist but no predictions): $R_{\text{diou}} = -0.8$.
- **False alarm** (predictions on a clean image): $R_{\text{diou}} = -0.3$.
- **Unmatched boxes**: each receives a penalty of -0.5 .

The final score is normalized by $\max(|G|, |P|, 1)$ and clipped to $[-1, 1]$.

Description consistency (R_{desc}). For each matched pair $(i, j) \in \mathcal{M}$, we compute the cosine similarity between the predicted and ground-truth descriptions using Qwen3-Embedding-0.6B. The raw similarity is linearly transformed to $[0, 1]$ via:

$$\hat{s}_{ij} = \text{clip} \left(\frac{\text{sim}(r_i, \hat{r}_j) - 0.5}{0.4}, 0, 1 \right). \quad (12)$$

The description reward is the sum of transformed similarities over matched pairs, divided by $\max(|G|, |P|, 1)$. Unmatched boxes contribute zero, implicitly penalizing over- or under-prediction.

Importance estimation (R_{imp}). For each matched pair $(i, j) \in \mathcal{M}$, the importance reward is:

$$r_{\text{imp}}^{(i, j)} = \text{clip} \left(1 - \frac{|\hat{s}_j - s_i|}{50}, 0, 1 \right), \quad (13)$$

where s_i and $\hat{s}_j$ are the ground-truth and predicted importance scores. This provides a continuous reward that decreases linearly with absolute error, reaching zero when the error exceeds 50 points.

D.2 GRPO Objective

The full GRPO objective optimizes the policy using clipped importance ratios and KL regularization:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E} [\min(\rho_s A_s, \text{clip}(\rho_s, 1 - \epsilon, 1 + \epsilon) A_s)] + \beta \text{KL}(\pi_\theta \| \pi_{\text{ref}}), \quad (14)$$

where

$$\rho_s = \frac{\pi_\theta(y_s | I, T)}{\pi_{\text{old}}(y_s | I, T)} \quad (15)$$

is the importance ratio for sampled response y_s , ϵ is the clipping range, and $\beta=0.01$ is the KL regularization coefficient.

D.3 SFT Hyperparameters

All generated images are resized so that the longer side is at most 1024 pixels. We use DeepSpeed ZeRO-2 with bfloat16 mixed precision on 16 GPUs. The learning rate is 3×10^{-5} with a cosine schedule and 5% warmup. We train for 1 epoch with per-device batch size 1 and gradient accumulation steps 1, yielding an effective batch size of 16; the $3 \times$ pre-baked jitter augmentation effectively exposes the model to three passes over each example within this single epoch. The maximum sequence length is 5,100 tokens, and the vision encoder is frozen throughout.

For coordinate jitter augmentation, each coordinate receives an independent random offset sampled uniformly from $[-10, +10]$ in the $[0, 1000]$ normalized space, with clamping to ensure valid box constraints. The offsets are resampled during SFT data loading across epochs, so the effective training corpus is not identical from one epoch to the next.

D.4 GRPO Hyperparameters

We use 16 GPUs with DeepSpeed ZeRO-2 and a learning rate of 5×10^{-6} . The KL coefficient is $\beta=0.01$. For each prompt, $S=8$ candidate responses are sampled via colocated vLLM rollout with temperature 1.0 and top- p 0.85 (max completion length 4,096 tokens). We train for 2 epochs with per-device batch size 4.

D.5 Compute Resources

All detector training experiments are run on GPUs. A single SDG detector SFT run takes approximately 2 hours on 16 GPUs, while a single detector GRPO run takes approximately 36 hours on 16 GPUs. For diffusion alignment, one BoxFlow-GRPO run takes approximately 24 hours on 16 GPUs. The defect-guided image refinement experiments use GPT-Image-1.5 through an external API; their wall-clock time depends primarily on API throughput and the allowed request concurrency rather than local GPU compute. The estimated wall-clock time for the reported experiments is about 7 days in total. The full research project required additional compute beyond this estimate because it included preliminary studies and failed experimental runs that are not reported in the paper.

D.6 Data, Code, and Model Availability

We provide code, model weights, reproduction instructions, and a sampled subset of SDG-30K at https://github.com/REPLACE_WITH_REPO. The complete SDG-30K dataset is undergoing release review and will be publicly released with documentation, annotation schema, data splits, and license and usage notices once approved.

D.7 Existing Assets and Licenses

We use existing assets only for research dataset construction, training, evaluation, or API-based editing, and cite their original sources throughout the paper. Pick-a-Pic prompts are used under the MIT License. Qwen3-VL-4B-Instruct and Qwen3-Embedding-0.6B are released under Apache-2.0. The T2I generators used for SDG-30K follow their respective public licenses or terms: FLUX.2-dev is governed by the FLUX Non-Commercial License, Z-Image-Turbo and LongCat-Image are released under Apache-2.0, and SANA-1.5 is released under NSCL v2-custom / NVIDIA License. Gemini 3 Pro, GPT-5.4, and GPT-Image-1.5 are accessed only through their official API terms. Public baselines and evaluation resources, including ImageDoctor, Flow-GRPO, UnifiedReward-2.0, PickScore, CLIPScore, HPSv3, DeQA, Forensic-Chat, DrawBench, and RichHF-18K, are cited and used according to their public releases or provider terms. We do not redistribute third-party weights, prompts, images, or code except where allowed by their licenses; released SDG assets will include attribution, license notices, intended-use documentation, and pointers to the original sources.

D.8 LLM Usage Declaration

LLMs and VLMs are used as core components of this work. Gemini 3 Pro is used during data preparation for description expansion, reasoning-trace distillation, and importance scoring; Qwen3-VL-4B-Instruct is fine-tuned as the SDG detector; Qwen3-Embedding-0.6B is used for description-similarity evaluation and reward computation; and GPT-Image-1.5 is used for defect-guided image

refinement. We also used general-purpose LLMs for manuscript writing assistance, including language polishing and wording refinement. All technical claims, experimental results, and final text were reviewed and edited by the authors.

E Extended Experimental Results

E.1 SDG Detector Ablation

Table 7 reports the full ablation results summarized in Section 5.1.

Table 7: Ablation study on SDG-30K test set. Table header follows Table 2. “–” indicates the component is ablated and the corresponding metric is undefined. Ablation groups are separated by dashed lines: training stage, CoT steps, output component, and architecture/augmentation.

Variant	Artifact						Misalignment					
	Image-Level		Defect-Level				Image-Level		Defect-Level			
	All	GT=0	GT>0				All	GT=0	GT>0			
	DerTypeF1	ClnAcc	BoxF1@0.1	BoxF1@0.5	DescCos@0.1	ImpAcc@0.1	DerTypeF1	ClnAcc	BoxF1@0.1	BoxF1@0.5	DescCos@0.1	ImpAcc@0.1
SFT	0.776	0.697	0.402	0.255	0.904	0.883	0.636	0.799	0.499	0.376	0.893	0.892
GRPO	0.772	0.560	0.404	0.263	0.904	0.887	0.675	0.732	0.511	0.387	0.888	0.893
SFT w/o step 1	0.719	0.793	0.357	0.218	0.894	0.880	0.597	0.798	0.436	0.329	0.875	0.892
SFT w/o step 3	0.731	0.537	0.387	0.253	0.897	0.884	0.496	0.840	0.364	0.281	0.846	0.878
GRPO (no think)	0.776	0.640	0.417	0.288	0.899	0.881	0.640	0.677	0.487	0.352	0.879	0.883
SFT w/o desc	0.757	0.718	0.386	0.250	-	0.884	0.609	0.782	0.468	0.358	-	0.894
SFT w/o importance	0.761	0.745	0.402	0.254	0.898	-	0.594	0.793	0.448	0.357	0.870	-
SFT unfreeze ViT	0.659	0.638	0.329	0.203	0.894	0.880	0.584	0.799	0.427	0.305	0.856	0.880
SFT w/o jitter	0.751	0.684	0.396	0.253	0.898	0.884	0.611	0.786	0.462	0.360	0.878	0.894

E.2 Extended Comparison with ImageDoctor on SDG-30K

Figure 7 extends the comparison in Figure 4 with six additional cases covering both *artifact* and *misalignment* defects, as well as images where the targeted defect type is absent. For ImageDoctor we show the artifact and misalignment heatmap heads separately; for SDG we overlay the predicted bounding boxes with per-instance labels. SDG consistently grounds prompt-conditioned misalignments (e.g., “Nucleosome” depicted as a double helix, “Dumbo” drawn without clown makeup) that ImageDoctor’s misalignment head misses, while avoiding the spurious face/hand activations that ImageDoctor’s artifact head produces on clean regions (rows 5–6).



Figure 7: Extended qualitative comparison on SDG-30K. Columns: original image, ground-truth SDG annotations, ImageDoctor artifact heatmap, ImageDoctor misalignment heatmap, and SDG (ours) predictions. Red text in the captions highlights the prompt span corresponding to the misalignment.

E.3 Full SDG Output Example

We provide the complete SDG model output for the example shown in Figure 1. The model receives the generated image and the prompt, and produces a structured reasoning trace (<think>) followed by a JSON defect set (<answer>).

SDG Full Output for Figure 1

Prompt: Psychedelic smoke, explosion, fire twirling, backlit, twisting, curled, petite American ballerina, wearing ballerina sparkling lace tutu, riding long glowing neon skateboard, star jump upside down, 8K, HD, highly detailed, rendered in octane

<think>

Step 1: Caption Understanding

- The caption describes a "petite American ballerina" in a "sparkling lace tutu" riding a "long glowing neon skateboard".
- Key action: "star jump upside down".
- Setting/Style: "psychedelic smoke", "explosion", "fire twirling", "backlit", "twisting", "curled", "8K", "highly detailed", "rendered in octane".

Step 2: Visual Analysis & Defect Spotting (Issue Summary)

- Misalignment (Action/Pose): The most prominent issue is the ballerina's pose. The caption explicitly requests a "star jump upside down". However, the generated image shows the ballerina standing upright on the skateboard, balancing on one leg with the other extended. She is neither upside down nor performing a star jump.
- Artifact (Anatomy - Hands): Both hands show generation artifacts. The hand raised high in the air (image-right) appears to have fused or indistinct fingers. The hand extended to the side (image-left) also lacks clear finger definition and looks somewhat malformed or "melted".
- Artifact (Anatomy - Face): The facial features, while detailed, show some subtle distortion typical of AI generation, particularly in the alignment of the eyes and mouth, which appear slightly uncanny or warped.

Step 3: Localization (Box-by-Box Grounding)

- Box 1 (Artifact): The defect is located on the raised hand on the image-right side; it is a small localized region where the fingers are visible but appear fused and lack proper anatomical separation.
- Box 2 (Artifact): The defect is on the extended hand on the image-left side; it is a small localized region at the end of the arm where the fingers are blurry and indistinct, failing to resolve into a clear hand shape.
- Box 3 (Artifact): The defect is centered on the ballerina's face; it is a part-sized region where the facial geometry (eyes, nose, mouth) appears slightly distorted and lacks the crisp symmetry expected of a high-quality render.
- Box 4 (Misalignment): The defect encompasses the entire figure of the ballerina; it is a large area covering the subject who is standing upright on the board, contradicting the prompt's instruction for an "upside down" and "star jump" pose.

</think>

<answer>

```
[
  {
    "box_2d": [600, 48, 641, 100],
    "label": "artifact",
    "description": "On the raised image-right hand, the fingers appear fused and indistinct, lacking clear anatomical separation and joint definition, resulting in a distorted, unnatural shape.",
    "importance": 22
  },
  {
    "box_2d": [223, 345, 285, 385],
    "label": "artifact",
    "description": "On the extended image-left hand, the fingers are malformed and blurry, showing a lack of distinct digits and a melted appearance rather than a properly articulated hand.",
    "importance": 22
  },
  {
    "box_2d": [184, 650, 688, 899]
```

```

"box_2d": [474, 220, 530, 281],
"label": "artifact",
"description": "The facial features in the upper center show
  subtle distortion, with the eyes and mouth appearing slightly
  warped or asymmetrical, failing to maintain a consistent
  realistic geometry.",
"importance": 32
},
{
  "box_2d": [223, 48, 657, 840],
  "label": "misalignment",
  "description": "The ballerina is depicted standing upright on
    the skateboard, which directly contradicts the caption's
    explicit requirement for an 'upside down' star jump action.",
  "importance": 95
}
]
</answer>

```

E.4 Extended BoxFlow-GRPO Qualitative Comparison

Figure 8 extends the qualitative comparison in Figure 5 with additional DrawBench prompts. Compared to FlowGRPO (UR2) and DenseFlow-GRPO (ImageDoctor), BoxFlow-GRPO (UR2+SDG) better respects fine-grained prompt attributes (e.g., correct color binding in “a red book and a yellow vase”, coherent subject composition in “Darth Vader playing with raccoon”) while preserving photographic realism, avoiding the illustration/anime drift that baseline RL variants exhibit.



Figure 8: Extended qualitative comparison of BoxFlow-GRPO on DrawBench prompts. Columns: Base (FLUX.1-dev), FlowGRPO with UR2 reward, DenseFlow-GRPO with ImageDoctor heatmap reward, and BoxFlow-GRPO with UR2+SDG structured reward (ours).

E.5 Extended Defect-Guided Refinement Results

Figure 9 extends the refinement comparison in Figure 6 with additional GPT-Image-1.5 editing cases. Across these examples, SDG feedback provides localized, instance-level guidance that helps the editor make targeted corrections, such as removing an extra lion cub when the prompt asks for a single cub, identifying an incorrect webpage title and replacing it with the correct title, and eliminating a large poster-like obstruction from a city-skyline image.



Figure 9: Extended qualitative comparison of defect-guided image refinement via GPT-Image-1.5. Columns: original image, Fixed caption-only editing, ImageDoctor heatmap/text-feedback editing, and SDG (ours) box-structured-feedback editing.

F Limitations and Broader Impact

F.1 Limitations

SDG currently focuses on two defect types, *artifact* and *misalignment*. Other quality dimensions, such as aesthetics, style, composition, safety, and cultural appropriateness, fall outside the current label space and would require additional annotation guidelines and evaluation metrics. The dataset is constructed from four contemporary T2I generators and Pick-a-Pic prompts; although this provides broad coverage, performance may change for other generators, domains, resolutions, or prompt distributions.

Our importance scores are distilled from Gemini 3 Pro under a fixed rubric. These scores provide a scalable severity signal, but they may differ from human preferences and may inherit biases from the teacher model. The SDG detector can also miss subtle defects, hallucinate defects in clean locations, or produce boxes that are too coarse for very small or highly diffuse failures. Finally, BoxFlow-GRPO assumes that box-derived penalties can be meaningfully projected onto latent spatial locations; this approximation may be less reliable when the latent grid does not align cleanly with visible defect locations.

F.2 Broader Impact

Structured defect feedback can make T2I systems more interpretable by exposing localized failure modes, supporting dataset auditing, and enabling targeted image refinement. These properties can help users diagnose generation errors rather than relying only on scalar preference scores.

The same capabilities also carry risks. Better diagnosis and refinement may improve the realism of synthetic images, which could be misused for deceptive or harmful content. Our release includes code, model weights, and sampled data, while the complete dataset is undergoing internal review before public release. Dataset and model releases should preserve annotation provenance, document intended use, and include safeguards for controlled release where appropriate. Because SDG can make localized judgments about generated people or scenes, downstream deployments should also consider fairness and bias in the underlying prompts, generators, annotators, and teacher models.